

# Kushwanth Parameshwaraiah

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## SUMMARY

Software Engineer with experience building high-impact, large-scale applications. Expert in Java (Spring Boot), Node.js, and microservices, with hands-on experience integrating Generative AI and RAG pipelines into production environments. Skilled in building secure, scalable, and user-centric web applications with cloud-native deployments using Docker, Kubernetes, Terraform, and monitoring with Prometheus and Grafana. Experienced in OAuth2, JWT, Spring Security, SQL/NoSQL databases and high-throughput messaging systems (Kafka, RabbitMQ).

## SKILLS

**Programming Languages:** Java, Python, TypeScript, JavaScript, SQL  
**Frontend Development:** React.js, Redux, React Query, Angular.js, HTML5, CSS3, TailwindCSS  
**Backend Development:** Spring Boot, Spring Cloud, Node.js, Express.js, Django, Flask, REST APIs, GraphQL, Microservices  
**AI/ML:** Generative AI, RAG Pipelines, TensorFlow, PyTorch, MLflow  
**Databases:** PostgreSQL, MySQL, MongoDB, Redis, Hibernate, JPA  
**Streaming/Messaging:** Apache Kafka, RabbitMQ  
**Cloud & DevOps:** AWS (EC2, S3, Lambda, EKS, IAM), Azure, Docker, Kubernetes, Helm, Terraform, CI/CD pipelines  
**Security:** OAuth2, JWT, Spring Security, RBAC, PCI DSS  
**Testing:** JUnit, Mockito, Jest, React Testing Library

## EXPERIENCE

### Cognizant Technology Solutions | Full Stack Engineer | August 2021 – January 2024

**Tech Stack:** React, Redux, Webpack, Jest, Cypress, Java (Spring Boot), Junit, Mockito, Docker, Jenkins, Kubernetes, MySQL

- Engineered scalable full-stack solutions for the Verizon POS application using React and Redux for centralized state management, Spring Boot, and Spring Cloud, driving a 25% increase in user engagement across 1500+ retail stores.
- Enhanced client-side performance via Webpack code-splitting and lazy loading, reducing Time to Interactive (TTI) by 40% and boosting user retention metrics by 15% for high-traffic sales interfaces.
- Increased system reliability by implementing 90%+ unit and integration test coverage using Jest and Cypress, which reduced production defects by 30% and ensured stable feature rollouts.
- Decoupled a legacy React monolith into a modular Micro-Frontend architecture, resulting in a 50% improvement in resource utilization and significantly faster independent deployment cycles.
- Implemented end-to-end CI/CD pipelines using Jenkins and Docker, automating build, test, and deployment workflows for microservices and frontend applications, reducing deployment time by 50% and minimizing manual errors.
- Partnered within an Agile/Scrum team to deliver high-quality features, utilizing Jira and Confluence for sprint planning, documentation, and cross-functional communication.

### Cognizant Technology Solutions | Software Engineer Intern | March 2021 – July 2021

**Tech Stack:** Angular, RxJS, Webpack, Jest, Java (Spring Boot), AWS, Junit, Docker, Jenkins, Kubernetes, MySQL, Redis

- Architected an enterprise-wide Audit Management Tool from the ground up, automating project tracking workflows and reducing manual audit processing time by 30% across multiple departments.
- Engineered robust and scalable RESTful APIs using Spring Boot and Express.js to support complex product workflows, engineering approval processes, and real-time vendor data integration with 99.9% uptime.
- Implemented highly responsive Angular dashboards with RxJS that provided intuitive, real-time visualization of product lifecycle stages, improving operational visibility and decision-making by 40%.
- Containerized microservices with Docker and deployed on AWS EC2 behind an Elastic Load Balancer, achieving 99.9% uptime and supporting horizontal scalability.
- Automated end-to-end CI/CD pipelines using Jenkins, Bitbucket and Docker Compose, reducing manual intervention and cutting deployment times by 50%.

## EDUCATION

**Masters in Computer Science and Engineering** | University at Buffalo SUNY, Buffalo, NY **June 2025**

## PROJECTS

**Kratos – Sales forecasting pipeline:** an e-commerce platform for games, designed to manage and distribute digital game products efficiently with focus on personalized sales forecasting for game publishers.

- Designed and deployed cloud infrastructure using Terraform (Infrastructure as Code) on GCP, enabling scalable and reproducible environments.
- Built ETL pipelines using Apache Airflow for automated data ingestion and processing, improving data pipeline efficiency by 30%.
- Developed scalable backend services (Node.js, Express.js) supporting high-volume transactions and concurrent users, improving system responsiveness under load.
- Integrated a sales forecasting ML pipeline (TensorFlow, MLflow) to analyze purchasing trends and improve demand prediction accuracy.

### AI Voice Scheduling Agent

- Engineered a high-concurrency voice agent using Retell AI, Python, and Node.js to autonomously manage Google Calendar scheduling via natural language processing.
- Designed low-latency backend services and API integrations (FastAPI, webhooks) to handle real-time voice interactions and event triggers.
- Architected custom LLM prompt pipelines to handle complex time-zone conversions, conflict resolution, and conversational state management, improving booking success rates.

### Norway Fishing Catch Prediction & Optimization

- Architected a predictive analytics engine using Python and XGBoost to classify fish species and preservation methods with up to 96.7% accuracy, utilizing automated feature selection to ensure high model reliability across high-dimensional oceanic datasets.
- Developed a modular training framework and automated ETL pipeline that incorporated translation, cleaning and PCA for dimensionality reduction, achieving a 15% F1-score improvement over baseline logistic regression models.
- Engineered a robust Django-based full-stack platform featuring a map-integrated GUI that provides real-time tool recommendations and species forecasting, effectively reducing manual equipment selection effort for expeditions.

## CERTIFICATIONS

- AWS Certified Developer – Associate (AWS Certified)
- Microsoft Certified: Azure Fundamentals